

Algebraic Topology: a beginner's course

This lecture:

**AlgTop32: Simplices and
simplicial complexes**

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Lecture 32 Simplicies + simplicial complexes



In A^3


$$\lambda_1 \vec{v}_1 + \lambda_2 \vec{v}_2 + \lambda_3 \vec{v}_3$$

where $0 \leq \lambda_i \leq 1$

and $\lambda_1 + \lambda_2 + \lambda_3 = 1$

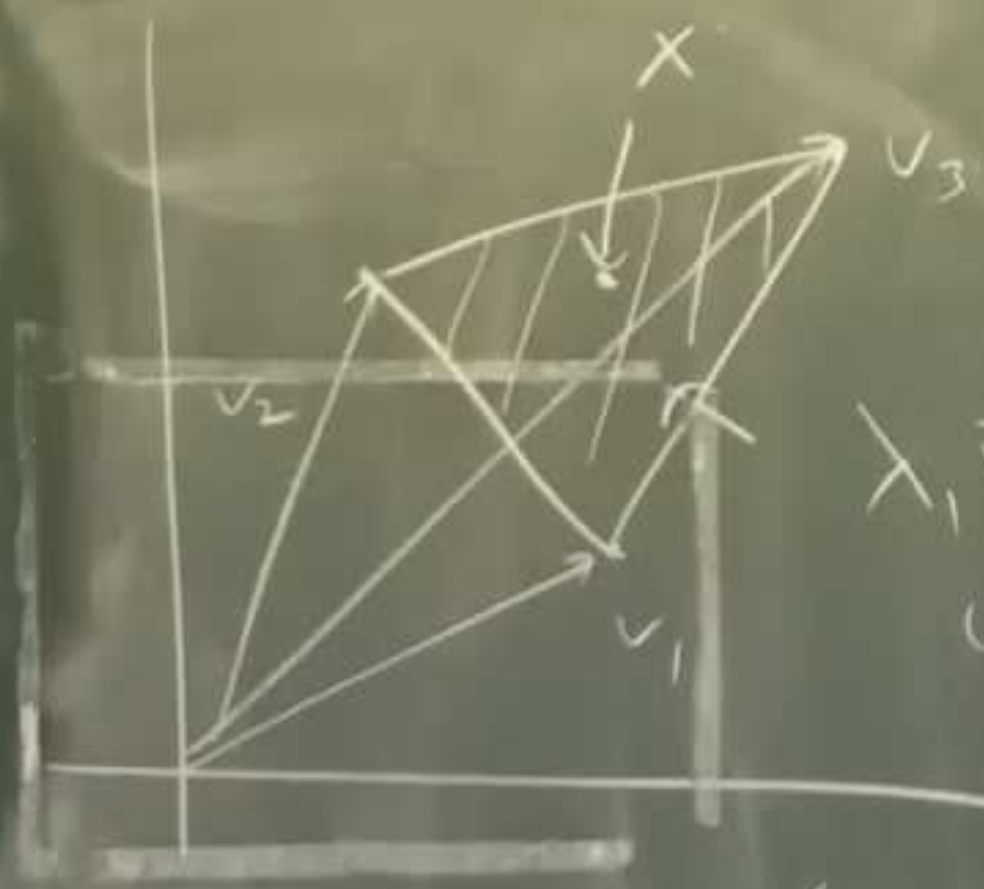
$(\lambda_1, \lambda_2, \lambda_3)$ called barycentric

co-ords for point in Δ

n-simplex Δ_n . $\lambda_0 \vec{v}_0 + \lambda_1 \vec{v}_1 + \dots + \lambda_n \vec{v}_n$ in A^3

$$0 \leq \lambda_i \leq 1$$

$$\lambda_0 + \lambda_1 + \dots + \lambda_n = 1$$



$$\lambda_1 \vec{v}_1 + \lambda_2 \vec{v}_2 + \lambda_3 \vec{v}_3$$

when $0 \leq \lambda_i \leq 1$

$$\text{and } \lambda_1 + \lambda_2 + \lambda_3 = 1$$

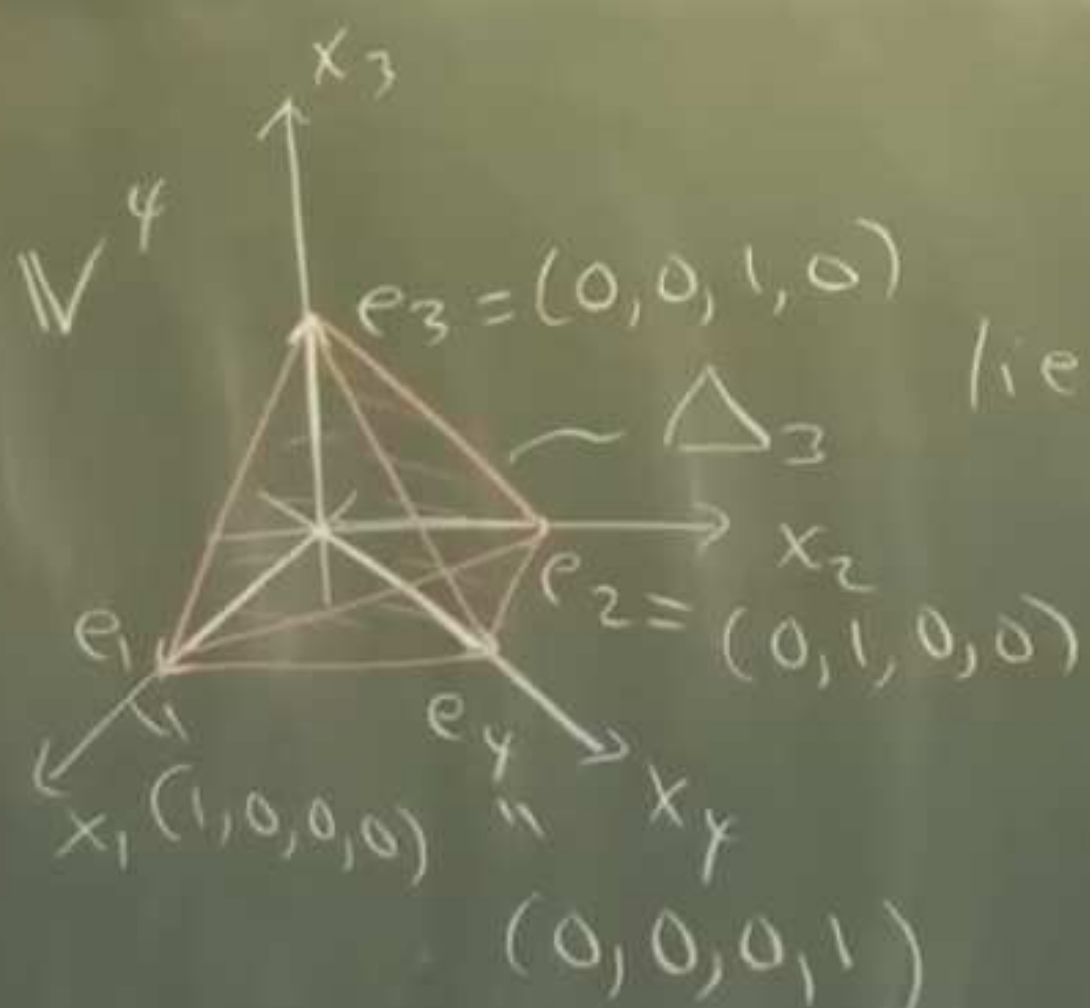
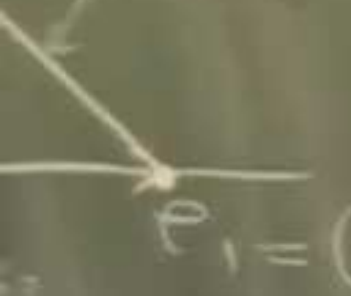
$(\lambda_1, \lambda_2, \lambda_3)$ called barycentric
co-ords for point in Δ .



in A^n . $\vec{v}_0, \vec{v}_1, \dots, \vec{v}_n$

Standard form for simplices

Δ_1 (line segment) V^4



lies in hyperplane

$$x_1 + x_2 + x_3 + x_4 = 1$$

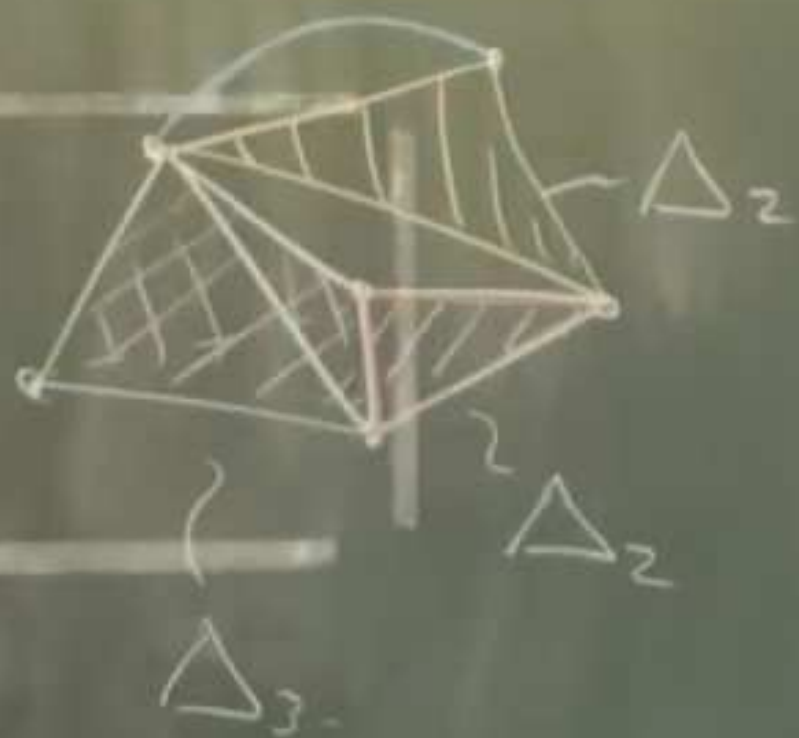
triangle

in plane

$$x_1 + x_2 + x_3 + x_4 = 1$$

Simplicial complexes

Build up a space using simplices



complexes

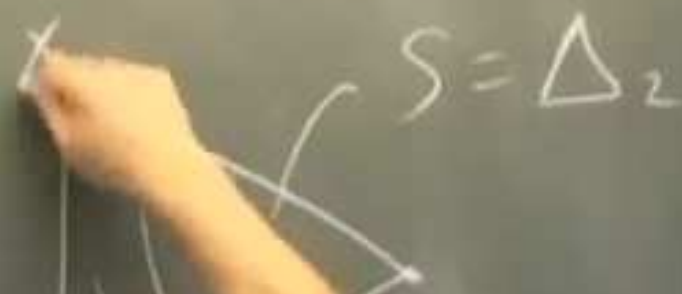
space using simplices



Any two simplices
are either disjoint,
or meet in a common
face.

Q. What is a face of a simplex?

A. Another simplex $P \subseteq S$ whose
vertices are also vertices of S .



simplexes

space using simplices

Any two simplices
are either disjoint,
or meet in a common
face.

Q What is a face of a simplex S ?

A Another simplex $P \subseteq S$ whose vertices
are also vertices of S .

